MINI BATTERY MANAGEMENT SYSTEM

A Simplified 3-Cell Battery Monitoring and Protection System

Presented by: VARSHA S

Embedded Systems Capstone Project 4 | Arduino Uno + Wokwi Simulation

Problem Statement

1. Develop a simplified BMS for a simulated 3-cell battery pack.
2. Monitor individual cell voltages and battery temperature continuously.
3. Detect overvoltage, undervoltage, overtemperature and cell imbalance.
4. Show a fault indication and move to a safe state when unsafe.

Objective

1. Monitor the voltage of each cell in a 3-cell battery pack.
2. Monitor the battery temperature.
3. Detect unsafe battery conditions.
4. Provide fault indication and safe operation (charge/discharge OFF).
5. Improve battery monitoring and protection.

System Inputs and Outputs

INPUTS

- Cell 1 voltage
- Cell 2 voltage
- Cell 3 voltage
- Battery temperature
- Charge / Discharge command

OUTPUTS

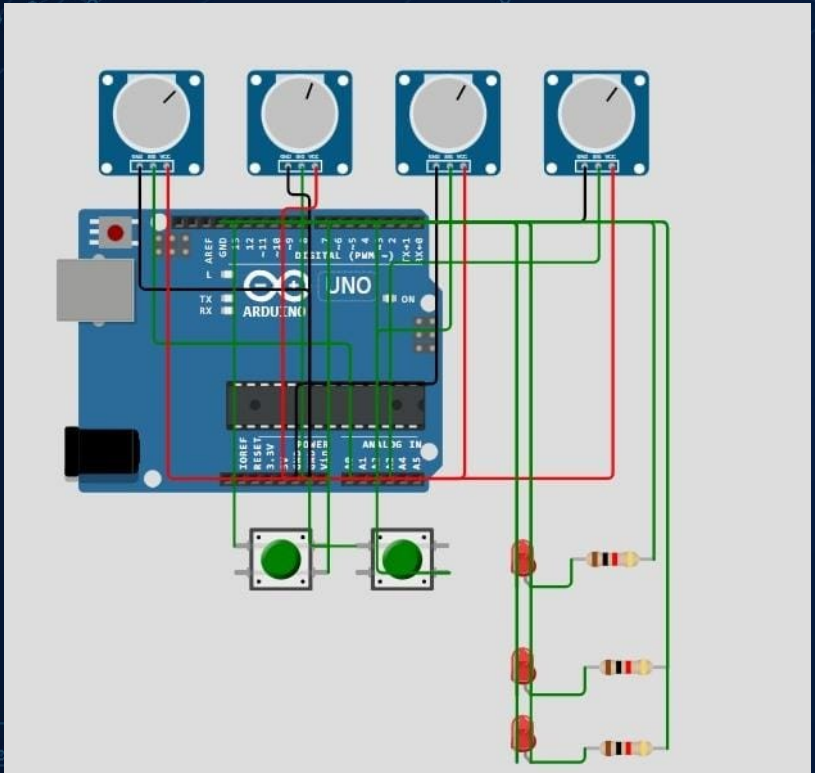
- Charge enable
- Discharge enable
- Fault indication
- Cell / Battery status
- UART diagnostic information

Concepts Used

ADC	Voltage Measurement	Filtering	SOC Estimation
Thresholds and Hysteresis	State Machine	Fault Management	Data Persistence

Concepts required by the project brief; core logic implemented: ADC reading, thresholds and fault management.

Wokwi Circuit Diagram



4 Potentiometers
Cell 1, Cell 2, Cell 3 voltage and temperature (A0 to A3)

2 Push buttons
Charge (D2) and Discharge (D3) commands

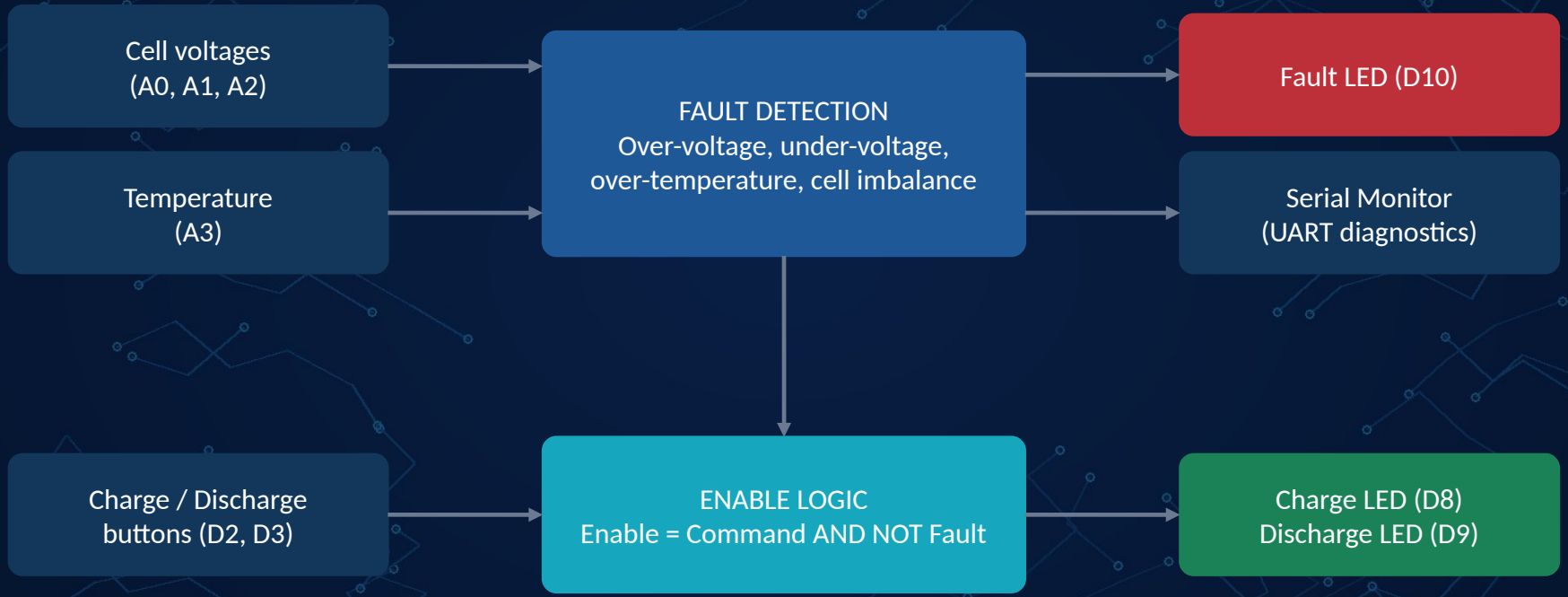
3 LEDs + resistors
Charge (D8), Discharge (D9), Fault (D10)

Arduino Uno
Reads inputs, decides, drives LEDs, prints to Serial

Pin Connections

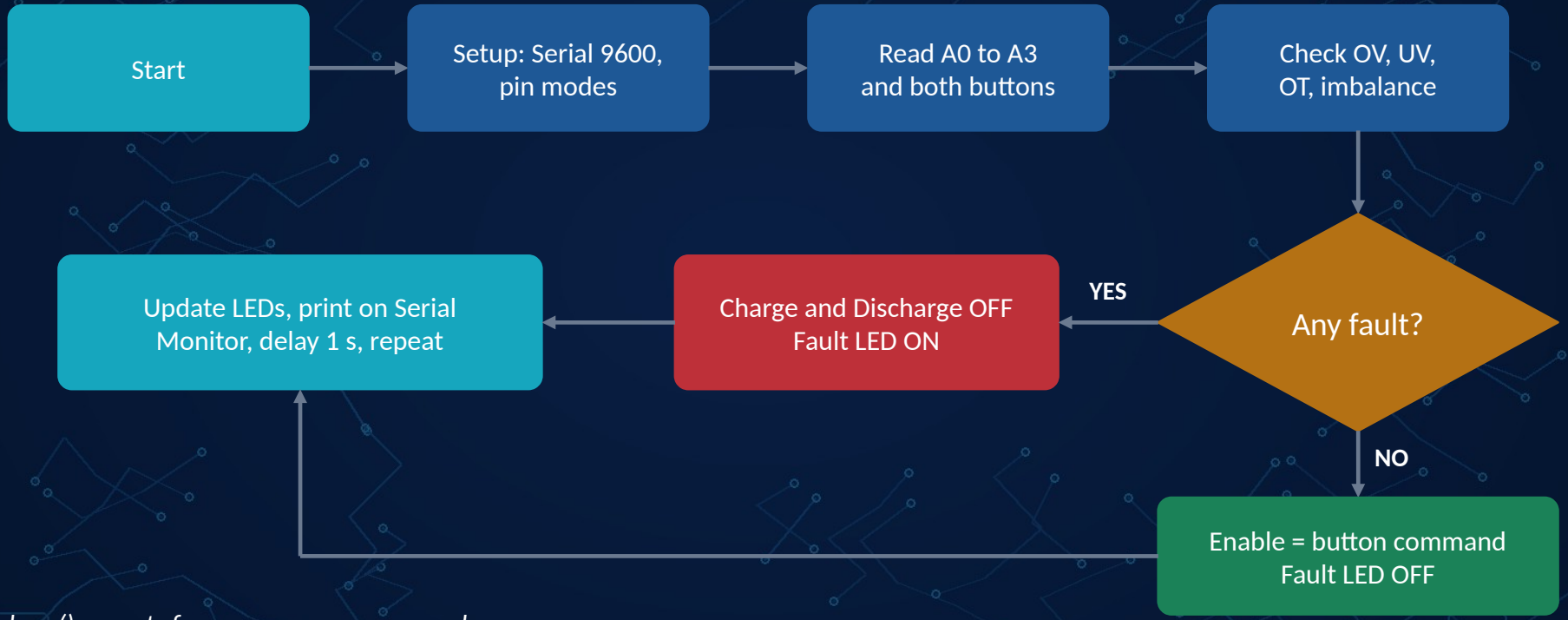
Signal	Component	Pin
Cell 1 voltage	Potentiometer	A0
Cell 2 voltage	Potentiometer	A1
Cell 3 voltage	Potentiometer	A2
Temperature	Potentiometer	A3
Charge command	Push button (INPUT_PULLUP)	D2
Discharge command	Push button (INPUT_PULLUP)	D3
Charge indication	LED + resistor	D8
Discharge indication	LED + resistor	D9
Fault indication	LED + resistor	D10

Logic Block Diagram



A fault always forces both Charge and Discharge to OFF.

Program Flow Chart



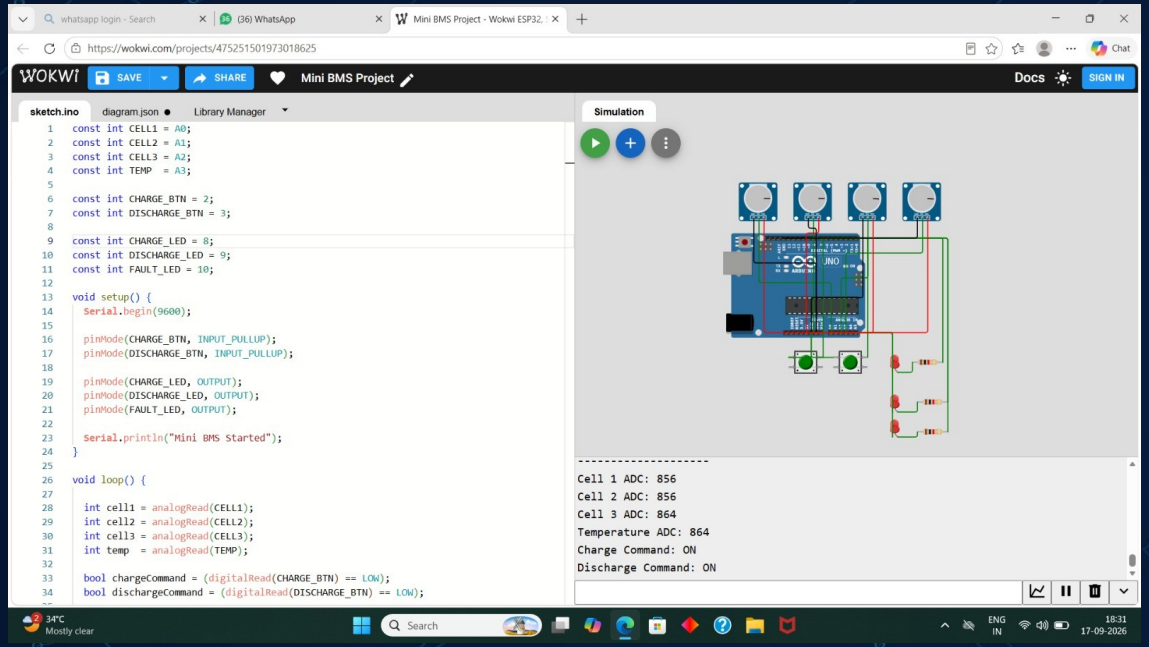
loop() repeats forever, once every second

Fault Logic and Thresholds

Condition	Rule in code (10-bit ADC, 0 to 1023)	Result
Over-voltage	any cell > 950	**FAULT**
Under-voltage	any cell < 500	**FAULT**
Over-temperature	temp > 900	**FAULT**
Cell imbalance	max cell - min cell > 100	**FAULT**

If any one condition is true, the battery is UNSAFE and both paths are blocked, even when a button is pressed.

Code Implementation (Arduino)

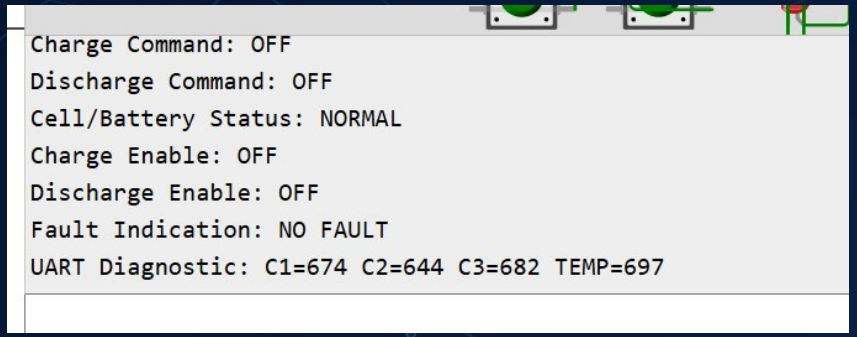


Screenshot: fault detection code and UNSAFE status on Serial Monitor.

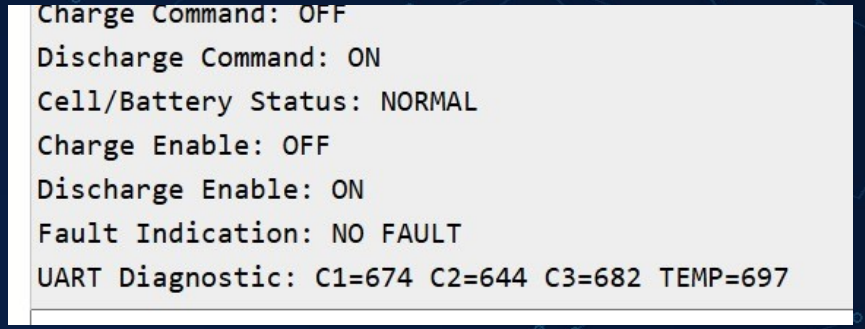
- analogRead() on A0 to A3 gives cell and temperature values
- INPUT_PULLUP buttons give Charge / Discharge commands
- Thresholds: cell > 950, cell < 500, temp > 900
- Imbalance: max - min > 100
- Fault: Status UNSAFE, both enables OFF

Simulation Output: Four Operating Cases

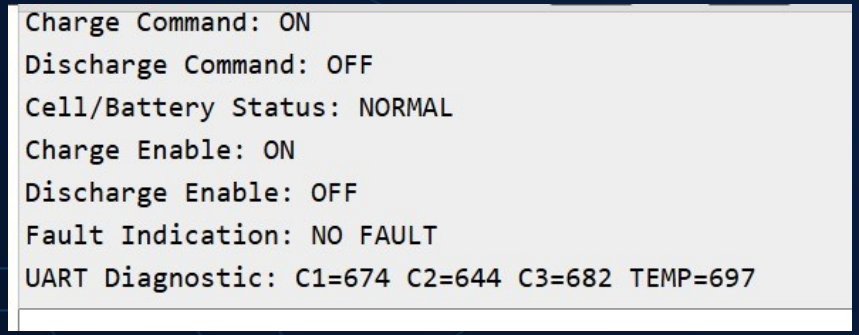
1. Idle (no command)



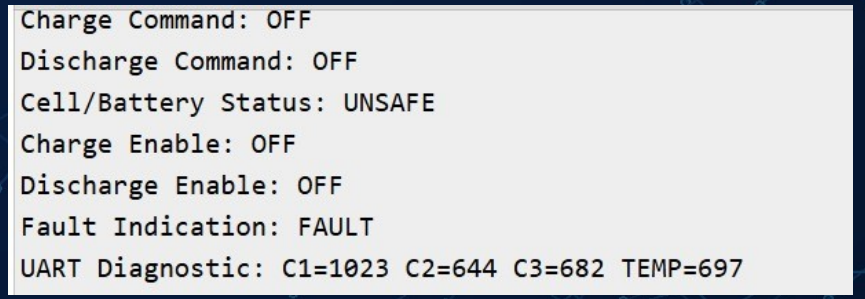
2. Discharge ON



3. Charge ON

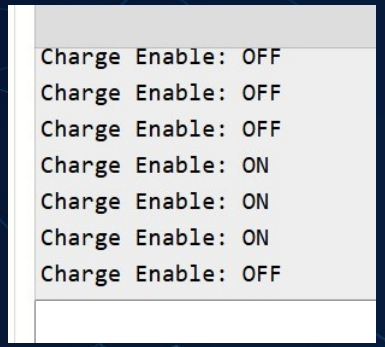


4. Fault: Cell 1 = 1023 (UNSAFE)

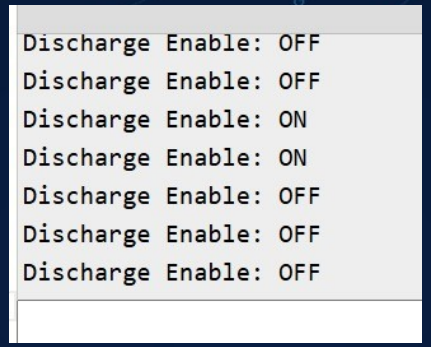


Serial Monitor Outputs

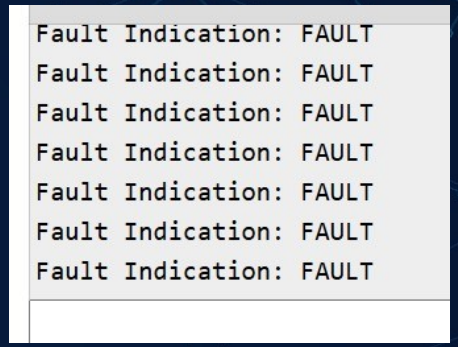
Charge Enable



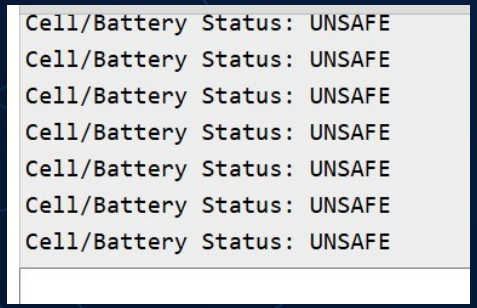
Discharge Enable



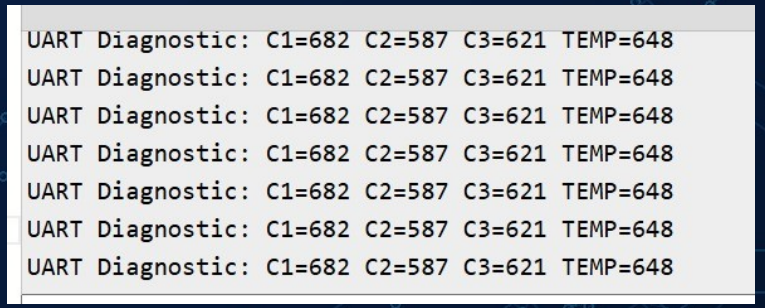
Fault Indication



Cell / Battery Status



UART Diagnostic



Development Stages and Fault Testing

Charge and Discharge enable logic with button and fault check

UART diagnostics and fault detection (OV, UV, OT, imbalance)

Integrated BMS: commands, status and all outputs together

Fault Testing

- Overvoltage condition tested
- Undervoltage condition tested
- Overtemperature condition tested
- System indicated FAULT for every unsafe condition

```
Fault Indication: FAULT
Fault Indication: FAULT
Fault Indication: FAULT
Fault Indication: FAULT
Fault Indication: FAULT
Fault Indication: FAULT
Fault Indication: FAULT
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Expected Outcome

1. Monitors 3-cell battery conditions continuously.
2. Detects overvoltage, undervoltage, overtemperature and cell imbalance.
3. Provides fault indication using LED and Serial Monitor.
4. Moves the system to a safe state (Charge and Discharge OFF).
5. Demonstrates a basic Battery Management System.

Conclusion

1. A simplified 3-cell BMS was implemented and simulated in Wokwi.
2. Cell voltages and temperature were monitored using Arduino ADC.
3. Fault conditions were detected and both Charge and Discharge were blocked.
4. The project demonstrates basic battery monitoring and protection.

THANK YOU

Mini Battery Management System | Presented by VARSHA S

Questions are welcome.